

KALPAN MUKHERJEE

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EDUCATION

Nitte Meenakshi Institute of Technology, Bangalore

2017 - 2021(Expected)

Bachelor of Technology in the Department of Information Science and Engineering

CGPA: 9.16

SKILLS

Areas of Expertise - Machine Learning, Computer Vision, Big Data, Networking

Technical Skills - C/C++, Neural Networks, Linux, SQL, React-Native, Python, Git, Distributed Systems

Frameworks - Tensor Flow, Scikit Learn, Numpy, GCP, OpenCV, Firebase, TCP/IP

EXPERIENCE

Research Intern @ Centre for Robotics Research, NMIT

October, 2018-May, 2020 (1y 8 months)

- Developed activity detection models for intruder alerts using auto-thresholding algorithms
- Compared effectiveness of Video Shot Detection in bright v/s poorly lit environments using image texture features like LAWS, GLCM and Texture Spectrum
- Produced short recaps of long border surveillance videos using clustering techniques

Technologies worked on - Linux, OpenCV, Thresholding, Filters, Textures, Feature Detectors

Software Engineering Intern @ Creatist

May, 2020 - July, 2020 (2 months)

- Developed a cross-platform mobile app for providing virtual classroom for institutes and corporates during the Covid-19 pandemic to assist with online learning efforts
- Built packages in React-Native to enable interactive classroom sessions by reworking features like Thumbnail, Video and Youtube iframe

Technologies worked on - React Native(CLI), Google Firebase, API integration

PROJECTS

ShowMeDaWae - A new routing map with road conditions @ VVCE, Mysore - Winners

TensorFlow with Keras, Google Maps API, React-Native — [Github](#) — [Youtube](#)

March, 2020

Helped Indian drivers navigate potholes while driving, by training a model classifying roads using Google Maps API to detect road conditions through accelerometer sensor data from a user's phone.

Electro Maps - Predicting electricity outage @ IIT, Dharwad - Finalists

TensorFlow with Keras, NodeJS — [Github](#) — [Medium\(Blog\)](#)

February, 2020

Predicting future electricity outage by deploying a model trained on past outage data scrapped from BBMP and BESCOM websites.

Handwriting Digitisation - Digitising classroom teaching

Linux, Arduino, Python, Google Cloud — [Youtube](#) — [Github](#)

October, 2019

Captured digital copies of handwriting on whiteboard using multiple arduino sensors working in unison. Custom smoothing algorithm is employed and final PDF is uploaded to Google Cloud. This helps digitise handwriting in languages where reliable OCR is not available.

ACHIEVEMENTS

Achievements - Winner at Onload 2.0 at VVCE-Mysore, Winner at Inspark-2019 at NMIT-Bangalore, Finalist at DevHack 2.0 at IIT-Dharwad

Blog Posts - [Predicting Electricity Outage](#), Deep dive into C++ STLs - [Unordered Maps](#), [Vectors](#)

Certification - Neural Networks and Deep Learning([Coursera](#)), Improving Deep Neural Networks: Hyperparameter tuning, Regularization and Optimization([Coursera](#)), Python([NPTEL](#))

Clubs - Core Team at [Pursuit of 42\(AI/ML club of NMIT\)](#), Member of [Hack Club, NMIT](#)